ABSTRACT

Road accidents are of grave concern in the world at large, resulting in innumerable killings and injuries annually. To detect this, scientists are resorting to the use of machine learning technology to forecast accidents even before they can occur. In this term paper, several machine learning strategies, including Decision Trees, Random Forest, and Neural Networks, are considered, which evaluate such factors as weather, traffic density, and road conditions and predict the possibility of an accident. The study provides a comparison of various methods of doing the research by way of reviewing 40 research papers with the aim of finding out which methods are most precise in their predictions.

The paper is going to provide a concise explanation of the role played by data quality and feature selection to enhance prediction accuracy. Numerous articles demonstrate that the inclusion of historical data of accidents together with up-to-date data, e.g., weather conditions and traffic, can improve the models. More sophisticated methods, such as deep learning, have demonstrated opportunities in recognizing intricate patterns, yet such issues as dataset bias and overfitting of the models have yet to be addressed. This part will present discussion based on these findings.

Also, the term paper will look at the significance of the location-based analysis in the prediction of accidents. The GPS and GIS integrated models can be used to determine the accident prone areas and thus the authorities can take preventive measures. But even though most algorithms are effective in simulation, when it comes to real-world applicability, their effectiveness depends on unpredictable conditions of the ways.

Lastly, this report is going to conclude the list of main findings of current research and give directions where in the future accident prediction systems could be enhanced. The comparison of various machine learning strategies will give the paper an informative idea of how to use technology to limit the number of accidents on the road and establish safety measures effectively. The results will be used as the basis of constructing an intelligent traffic control system, which is smarter and more responsive.

Keywords - Road Accident Prediction, Machine Learning, GPS, Deep Learning.

INTRODUCTION

Every time when we get into a car or cross a road, we are making a minor yet existing risk. Road accidents do not only run on the front page they break families, communities, and lives. Suppose we know when these disasters will occur, like when we know when there will be a storm and we know not to sail in dangerous weather. This is precisely what modern technology is attempting to do via super computer systems that learn the lessons of old accidents in order to ward off occurrences of the same in the future.

Machine learning - let us discuss that, it is an analogy, imagine you are coaching a small child by feeling. Much in the way a child learns to avoid a hot stove by being burned once, computer programs by examining thousands of accident reports can observe dangerous trends. Such digital detectives do not memorize these facts, they build up a sort of intuition as to which combination of weather, road surface conditions and behavior of the driver play into a disaster.

Think about decision trees as a 20 questions game. The computer poses such questions as: "Was it a night time?" Did the road get wet? The end points of each answer are only to find a conclusion concerning the risk of accidents. However, as much as we cannot depend on the view that one person gives out, we cannot also depend on a single tree. This is why we build entire forests of them,\what scientists (mistakenly) refer to as Random Forests nowhere near as random, in fact, as electronic trees used in classrooms a century ago in which droves of electronic trees cast their vote on the most safe conclusion.



The way our brains operate is by synapsing billions of neurons and researchers developed replications of the same thing called neural networks. Such computerized minds could intuitively notice intricate threats that a lower-order system would not pick up on - such as the fact that the pairing of fog and rush hour traffic conditions and a sharp bend create an ideal environment in which accidents take place. The more network layers deep learning is given (the more layers these networks have) the more intelligent it is to notice such subtle clues of danger.

However, the most intelligent system requires the proper information. Deciding on what facts to use - to pick and to choose is feature selection as the engineers call it - is akin to packing before a trip. Transport too much and you are slowed; leave anything important behind and you are in trouble. Another challenge we have to contend with is that accidents are mercifully lower (or so one should hope) than safe trips, making it more difficult to forecast - they are among the needles in the haystack of ordinary traffic data.

Modern systems have the ability to make danger zone heat maps using digital maps (GIS) and overlying satellite tracking (GPS) technologies. Try thinking of the navigation in your phone telling you of the traffic jams, only this time it is blinking at you with messages such as: "Accident hazard in the next 5 miles - please use alternative route." Experimental systems can go to the extent of watching real- time live weather radar and social media postings about road conditions to make real-time predictions.

It is not actually a technological issue but rather about guarding citizens. Knowing how these digital lifeguards do their job, we can improve them in doing their life-saving work. The eventual aim is not to make perfect predictions (though we shall continue in this effort) but rather to provide passengers, urban designers and rescuers with superior instruments to ensure our streets are safe. After all, the more accidents that are prevented the more people are making it home safely tonight.

Structure of the Paper

This paper will start with the Literature Review section, during which I will evaluate current literature available on the subject of prediction of road accidents based on machine learning. Both studies are summarized with the main particularities such as the name of an author, the year of the publication of the research, the methodologies applied, and limitations. This chapter gives an exhaustive basis of comprehending contemporary developments and drawbacks in the discipline.

In the Methodology section, I consider some different machine learning methods used in the area of accident prediction, including Decision Trees, Random Forest, and Neural Network. I mention the processing of data, such as weather conditions, traffic flow, and road features that can be used to anticipate high-risk scenarios by these methods. They contain practical examples and studies in order to demonstrate how effective they can be and how practical they are in the real world.

Analysis & Conclusion sections conclude by summarizing the advantages and drawbacks of various options, also evaluating the limitation issues such as data imbalance or high demand of the real-time processing. I offer future research directions and potential solutions that would increase precision of prediction. The paper concludes with references, that is, all the works used in the paper are mentioned.

LITERATURE REVIEW

Author[1] This author studied the data of accidents that happened in the UK in 2005-2014; they considered the severity of crashes, injuries, and the vehicles used. To enhance accuracy they cleaned the data by eliminating errors and abnormal entries. The tests were conducted on 4 models of predication: decision tree, random forest, logistic regression, and naive Bayes.The best results were observed on random forest, with 85 and 84 percent accuracy on severity and casualty, respectively. Naive Bayes was not successful since assumptions were too simple in case of real-life accident statistics. Results indicated that incidents were severe under poor lightning, fog, or flooded streets. The Southeast of the UK experienced the highest number of crashes but the safety measures have reduced the crashes with time. As compared to most studies, this study involved the comparison of several models simultaneously in order to provide more favorable outcomes.They employed the services of Apache Spark, a rapid data-processing program, to address big parameters of data.Cities can plan better with the help of the research: they can repair dangerous roads or position ambulances in strategic places.It also implies alterations to data approaches in order to make predictions on rare but fatal mishaps.It could have an actual application, such as focusing on dangerous districts to respond more promptly to emergencies. Untapped enhancements may incorporate the weather condition such as speed of the wind to enhance forecasting.

Author[2] This paper aims at predicting the severity of road accidents, analyzing the factors that contribute to it with the road accident data in New Zealand (2016- 2020) as a case study, using explainable machine learning (ML) models. The current study compares the performance of the ensemble ML models, Random Forest (RF), Decision Jungle (DJ), AdaBoost, XGBoost, LGBM, and CatBoost, that would be useful in the categorization of the accident severity locations into the following classes: either fatal, serious, minor, or non-injury. RF became the most suitable model giving it 81.45 of accuracy, 81.68 precision and 81.04, F1-score. One of the peculiarities of this work is that it uses Shapley Additive Explanations (SHAP) to explore interpretability, and the results show that road type and whether the collision involves multiple vehicles have major consequences on the severity of the injury. SMOTE is also used in the study to deal with the class imbalance and retrains models on features ranked high, which results in performance enhancement up to 8%. Key findings indicate such important predictors as drug/alcohol involvement, road geometry, and age of the driver. The study also offers practical guidelines that policymakers should use to improve road safety policies, which fall in line with the Road to Zero vision in New Zealand. Shortcomings relate to the specificity of data sets and the applicability of SMOTE with high dimensions data implying further study to include the use of deep learning in the management of imbalances.

Author[3] The proposed study aims at examining the available data on traffic accidents occurring in the Setbal district in Portugal (20162019) with the help of machine learning to better understand aspects that affect the severity of accidents and where accidents occur. The authors have used supervised (decision tree, random forest, logistic regression, naive Bayes) as well as unsupervised (DBSCAN, hierarchical clustering) methods. The major findings included severity of the motorcycle/pedestrian accidents in urban settings and hit and run accidents were not most likely to endanger any injuries. The factor that made this research unique was the combination of clustering (DBSCAN) and rule-based (C5.0 algorithm) modeling to provide comprehensible rules of severity of accidents. In hotspot prediction, GPS coordinates and weather data were utilized by the team to establish negative samples in the effort to balance the data. Other models were outperformed by random forests that resulted in the 73 percent accuracy with minimal false positive rate (3 percent). Another factor which the study emphasized on was the significance of real-time data fusion (weather, traffic, road conditions) to the predictive accuracy. The value of the work is that it provides a road safety study that can be used to take action by governments and drivers which will in future be extended to include other districts and deep learning models. The study was confirmed by the professionals in the sphere of road safety, and this fact highlights its practical value.

Author[4] The proposed research paper involves the use of machine learning with an aim of analysing and predicting severity of road accidents to help in reducing the number of fatalities and injuries. The authors also discuss the global implication of road accidents with an average death of 1.25 million every year, which stresses the necessity of data-driven resolutions. Machine learning algorithms allow defining the severity of accidents using various factors such as weather, road conditions, and traffic patterns: K-Nearest Neighbors (KNN), Naive Bayes, and Decision Trees are used in the study. This is done by pre-processing the data, selecting features of the data and training the model with historical datasets measuring accidents. KNN algorithm classifies accidents based on the proximity, Naive Bayes counts the probable events based on the Bayes Theorem, and decision trees divide the data in form of hierarchy to predict it. The architecture system has upload and visualization user interfaces and real-time data analysis that can be used by policy makers. As the main peculiarity of this study we could note the combination of several algorithms to compare their accuracy levels and Decision Trees appeared to demonstrate better results in severity predictions. A future mobile app is also suggested in the paper in order to provide real-time warnings to road users. The study will help to understand accident-prone locations and core factors to work out in advance and manage the traffic better to minimize the casualties. The results highlight the case of machine learning to enhance road safety approaches worldwide.

Author[5] This research paper aims at the use of machine learning (ML) to estimate the severity of road accidents to improve traffic safety and increase the efficiency of future emergency response plans. A broad variety of significant determinants is studied: the property of the vehicle (e.g. vehicle type, speed), road aspects (e.g. the type of surface, weather), driver-related factors (e.g. age, experience). A 12,316 record of accidents data were preprocessed by maintenance of missing values and categorical variables and then split into 80-20 ratio for training and testing.
Five ML algorithms, Decision Tree, Random Forest, K-Nearest Neighbors (KNN) Naive Bayes, and Adaptive Boosting (AdaBoost) were used and tested. Among them, Naive Bayes and AdaBoost were the most accurate in terms of predictions (85.3 percent). The research is also unique in that the trained model is deployed into a web-based app with the use of Flask, HTML and CSS that can provide real time prediction of the severity of a given traffic condition-this can also be an input to traffic management systems. This was observed when visual analysis involving correlation matrices and Bar charts was used to reveal very important insights such as more ferocious accidents occurring among the 18-50-year-old drivers during the daytime and "Dry" road conditions that were common to the majority of accidents. The article highlights the need to have high quality, comprehensive datasets in ensuring reliable models and raises such ethical issues as data confidentiality and bias of algorithms. To enhance the accuracy of their predictions, the authors offer to add real-time traffic and weather information along with more complicated methods such as deep learning in the future. The pitching point of this research is an all-in-one nature it has, i.e. the combination of strong ML techniques with a user-friendly web-based interface, which makes it a desirable asset to the work of policymakers and first responders to counter the problem of road accidents.

Author[6] The aim of this study entails predicting the severity of the road accidents through machine learning with the aim of increasing road safety. The authors have concentrated on the increase in accidents all over the world with India playing a significant role as such reasons and rationales to increase accidents include speeding, drunk driving, and bad roads that increase the problems. Three algorithms, including Logistic Regression, Decision Tree, and Random Forest are applied on a data set consisting of 1.6 million records of accidents (20052015) to predict the severity of the accidents. The major points were data preprocessing (cleaning, normalization, and visualization), feature analysis (weather, road conditions, driver age) and the training of the model. What was unique to the study is the fact that it includes hyperparameter tuning to get the model performance at its best, which is often neglected in other studies of the same kind. Random Forest was able to reach the maximum percentage of accuracy (86.86 percent), preceding Logistic Regression (86.23 percent) and Decision Tree (75.26 percent).Visualizations also disclosed the trends, including the fact that accidents were the highest on Thursdays and the drivers aged between 25 and 35. This analysis of correlation has determined the speed limits and urban/rural areas as an important factor. The conclusion of the paper is that Random Forest represents the best severity prediction model and can be incorporated in the future with other systems available in real-time to prevent accidents like Google Maps. The strengths of this

research are the combination of a broad dataset analysis and hyperparameter tuning, which provides the large-scale capability that can be applied by the traffic management authorities.

Author[7] In this study, an ensemble model, RFCNN, based on Random Forest (RF) and Convolutional Neural Network (CNN) has been proposed that aims at predicting traffic accident severity. This study covers the issue of road accidents that have become a priority in the developing world due to the high numbers of casualties and the economic loss incurred because of the accidents. With the help of RF, the authors took a US accident dataset (20162020) and revealed 20 important features, including distance, temperature, and humidity. Two stages of the methodology will be applied; in the first one, all 48 features will be used and in the second stage only 20 best features are used to train models. The RFCNN suggested incorporating a soft voting mechanism to combine RF and CNN predictions further improving the decision making. As an example, when RF predicts [0.6, 0.2, 0.7, 0.4] and CNN predicts [0.5, 0.4, 0.8, 0.5]) RFCNN averages to choose the highest probability class (in this example, class 3 with 0.75). The findings indicate that RFCNN performs better than base models (RF, AdaBoost, etc.), recording accuracy, precision (0.991, 0.974), and recall values (0.996) when only the top 20 features were used. An interesting point is that the fusion is performed at the level of the decision between machine learning (RF) and deep learning (CNN) approaches with a demonstration of strengthened features. The research underlines the effect of feature reduction enhancing performance with decreasing data collection costs. Such a strategy can help policy makers to focus on critical determinants of accidents which can up-scale accordingly in ensuring traffic safety. Extending RFCNN to cover multi-domain data sets is a possibility of future work.

Author[8] The main objective of this research paper is to develop a predictive model of road accidents in India with the help of high end data mining techniques. It pays attention to the data about accidents in Bangalore (2014 2017) in search of discovering hidden patterns related to risky road conditions or hotspots on speeding and integrates the Apriori algorithm of discovering patterns with naive Bayes and SVM related to risk classification. The model works on artificial neural networks (ANN) unlike in the traditional methods which allows addressing complex, non linear relationship of accidents data, enhancing prediction accuracy. Some of the key findings include the fact that more people die in rural regions as opposed to urban regions and that young men (20-35 years) are the most at-risk group since they drive carelessly. The weather, design of roads and type of vehicle are also important factors highlighted in the study. The team cleaned noise and normalized raw data of police reports and FIRs through the preprocessing of the raw data. Interactive R-tools were used to present trends, i.e., the increasing number of heavy-vehicle collisions, whereas the user-friendly web interface used to display the model enables the officials to feed the new accident data and update the trends on a real-time basis. This method not only identifies the high risk areas but is able to recommend safety action, such as tougher policing of speed at the junctions. A combination of rule mining (Apriori) and machine learning

(SVM/ANN) is a scalable solution to developing countries where there are few records of accidents. Finally, the model will help policy designers manage to save lives. It will support policymakers in making decisions concerning prioritizing infrastructure improvement and raising awareness on the basis of evidence-based data.

Author[9] The given study assesses how well machine learning (ML) models that involve the problem of road accident severity in Thailand predict fatalities and non-fatalities. The study fills a void in the literature of the comparative studies on Ml algorithms based on large scale crash data. There are eight supervised ML models, including Decision Tree, SVM, Random Forest, kNN, Neural Network, Naive Bayes, Logistic Regression and Gradient Boosting, trained on the dataset of 112,837 accidents (20152020) including at-fault drivers. Data preprocessing was done, in missing values, feature selection and class imbalance. The best-performing model was Random Forest with a good average AUC of 0.768, accuracy of 0.777, precision of 0.752, and recall of 0.777. Nevertheless, the recall of fatalities was limited (0.198) which shows that it is difficult to predict fatal crashes. Important predictive characteristics were driving on the highway, speeding, acting during daylight hours, not having any light at night, and being a male. The research conducted a unique comparison of various ML algorithms in performing a binary classification performed by multiple ML algorithms and their strengths and weaknesses.AUC, precision, and recall were found by utilizing confusion matrices. The analysis placed a lot of value on feature engineering and the most influential features were highway and gender. Accurate prediction of non-fatal accidents was performed, but low number of correctly recalled fatalities implied using more complex methods such as SMOTE or the ensemble approach. The results indicate that ML can be used in road safety, whereas better fatality prediction models are needed. Proper lawfulness and improved road illumination are some examples of targeted attainment which the policymakers can employ using these insights. The thorough model comparison and large-scale analysis of a dataset are among the strengths of the study that provides a guide on moves in the further research of the area of accident severity prediction.

Author[10] The aim of the study is to examine the performance of commonly used machine learning classifiers to predict road traffic accidents (RTAs) based on real world data of the Gauteng region in South Africa. The study seeks to support the works of the policymakers and the transport authorities by helping them know the most effective accident severity predictive models. Six classifiers (naive Bayes, logistic regressions, k-nearest neighbor, AdaBoost, support-vector machine, and random forests) were tested together with five imputation methods of missing value (mean, median, k-NN, and MICE) and reduction of dimensionality (PCA and LDA). The approach used was to preprocess the data, deal with missing values, and fit in both default and optimized classifiers. The performance was measured in such metrics as accuracy, RMSE, precision and recall, and the AUC-ROC curves. A model that performed best was the use of random forest (RF) classifier, which was supplemented with MICE to resolve the missing information, including 97 percent accuracy and having lowest root mean square error (RMSE),

which is 0.01. The dimensionality reduction showed that LDA had more effective results to use in multi-class classification as compared to PCA.The particular feature of this study was the complete comparison of methods of imputation of missing data, during which the best method was found to be MICE in maintaining data integrity. The results stress that RF is a solid tool in RTA prediction, and that giving consideration to them may be helpful in enhancing road safety policies. The study supports global goals of Sustainable Development with reference to the high RTAs fatality rates especially in the low- and middle-income countries.

Author[11] The aim of the study is to examine the performance of commonly used machine learning classifiers to predict road traffic accidents (RTAs) based on a real world data of the Gauteng region in South Africa. The study seeks to support the works of the policymakers and the transport authorities by helping them know the most effective accident severity predictive models. Six classifiers (naive Bayes, logistic regressions, k-nearest neighbor, AdaBoost, support-vector machine, and random forests) were tested together with five imputation methods of missing value (mean, median, k-NN, and MICE) and reduction of dimensionality (PCA and LDA).The approach used was to preprocess the data, deal with missing values, and fit in both default and optimized classifiers. The performance was measured in such metrics as accuracy, RMSE, precision and recall, and the AUC-ROC curves. A model that performed best was the use of random forest (RF) classifier, which was supplemented with MICE to resolve the missing information, including 97 percent accuracy and having lowest root mean square error (RMSE), which is 0.01. The dimensionality reduction showed that LDA had more effective results to use in multi-class classification as compared to PCA.The particular feature of this study was the complete comparison of methods of imputation of missing data, during which the best method was found to be MICE in maintaining data integrity. The results stress that RF is a solid tool in RTA prediction, and that giving consideration to them may be helpful in enhancing road safety policies. The study supports global goals of Sustainable Development with reference to the high RTAs fatality rates especially in the low- and middle-income countries.

Author[12] The given research paper is a survey conducted by S. Vasavi (2021) titled A Survey on Extracting Hidden Patterns within Road Accident Data using Machine Learning Techniques which centers around analysis of the road accident data to find the hidden patterns and minimize the number of accidents. The paper is based on the data available in Krishna district national highways (2013) and employs the machine learning methods of clustering, association rule mining etc. Such key factors as emotions of the drivers, weather conditions, traffic conditions, type of the road, and speed of car movement are investigated.The methodology implies preprocessing of the data (cleaning, transformation), analysis of its relevance and visualization through density histograms that emphasize such correlations as fatal accidents vs. weather, time or age. The peculiarity of this study lies in the fact that instead of using either clustering or association rule mining, it provides the combination of both these methods to draw the actionable insights, including the higher accident rates during weekends, colder nights, and among older

drivers.All results indicate that accidents occur the most in August of which males are involved in 73.45% of accidents. Risks can easily be communicated to the policymakers through visualization aids such as histograms. The paper is closed by proposals of future research involving more profound data and complex methods, such as deep learning. Such a strategy is distinct in terms of combining several types of machine learning and performing an effective visualization to support road safety decisions.

Author[13] This report makes a comparison of statistical and machine-learning approaches to evidence-based classification of the severity of road traffic accidents in the area of Setubal district, Portugal, with the data of 2016 19. The study deals with the high mortality rate in Portugal by examining two models which are severity model (accidents involving victim and damage to property) and serious injuries model (fatal/serious injuries and minor injuries). It utilized logistic regression and machine-learning models (Random Forest, Naive Bayes, SVM, KNN and C5.0).Despite the large imbalance in the data to compare logistic regression with machine-learning models (e.g., serious injuries model with 10:1 imbalance), logistic regression still performed especially better than the machine-learning models (e.g., serious injuries model). This study is unique since a spatial analysis in grouping municipalities in terms of accident severity has been employed thereby reducing the number of variables and raising the efficiency of the model.Methodologically, the article put more focus on data-preprocessing (e.g., removing missing values, but not imputing them), including variable transformation to fit a logistic regression (e.g., converting continuous factors: age, the number of the driver). Measurements of performance such as sensitivity, specificity, and G-mean indicated the superiority of the logistic regression in terms of interpretability, where it provides odds ratios to determine risk factors (e.g. the increase of severity during the weekends, motorcycle riding, or unpaved roads).The research findings suggest that statistical models should be used as the active minority groups in the research because of their explanatory capabilities to explain models hence, the suitability of machine-learning models to be applied in bigger datasets. Further research will generalize the analysis on other areas and find deep-learning methods to detect the rare events.

Author[14] The present research paper proposes a machine learning framework to process road accident data so that it could minimize the number of victims and ensure better road safety. The research paper raises the concern of the increased number of road accidents in the world and the forecasts predict an upcoming 8% jump in the casualties by 2030. The authors apply the methods of unsupervised learning, specifically, the K-means clustering to divide heterogeneous Toronto accidents data (20072017) into high, moderate, and low-frequency zones. Not unique to the present paper is the combination of feature selection and Parzen Probabilistic Neural Network (PPNN) towards the pulling of hidden patterns where 85 percent accuracy was found in the prediction of critical situations.In terms of methodology, the paper is about data preprocessing (missing values cleaning, encoding), exploratory data analysis (visualization on a monthly, annual, regional basis). The most important points to note are damage caused by aggressive

driving (62.9 percent), speeding (21.4 percent), and the east region in the city of Toronto being the most likely area to have accidents. The K-means clustering based the data on four clusters in order to determine the risk areas and feature selection set the optimal modeling by facilitating the least significant attributes such as accident numbers or minute-level timestamps.The paper concludes that machine learning can be used to locate accident hotspots and contribute factors and, thus, assist policymakers in specific intercessions. The proposed method based on clustering of segmentation and PPNN based pattern detection provides a distinctive feature to the road safety study in terms of patterns detection incorporated in the analysis in contrast with previous studies based on conventional statistical techniques.

Author[15] The research was conducted on the formulation of crash severity on vulnerable road users (VRUs) which entailed the pedagogical use of machine learning models to determine the main risk factors. The study used crash data from Queensland, Australia (20132019) that includes 21,158 incidents and 17 features, such as road conditions, speed limits and time of day. Each of the VRU groups was studied and combined in three machine learning models through training and testing Random Forest (RF), Support Vector Machine (SVM) and K-Nearest Neighbor (KNN). Part of the distinctiveness of this study was in that it compared the feature importance and partial dependence plot and presented how certain factors varied in severity in different VRUs. Preprocessing of data was done by binary categorisation of injury severity (low vs. high), in order to overcome the imbalance issue during training of the model, numerical codes were used. The optimal model was RF which obtained accuracies of 72.3 per cent, 64.45 per cent, 67.23 per cent and 68.57 percent of motorcyclists, bicyclists, pedestrians and unified VRUs respectively. The prominent revelations included the most vulnerable risk of motorcyclists especially regarding the speed limits and time of day and the lowest risk of the bicyclists. Predictors critical were the "hour of crash" and the "posted speed limit" and odds and ends were the driver age group. Partial dependency analysis that was used in the study was a rare visualization that demonstrated how the changes in the feature, like an increase of speed limits, lead to a rise in probability of damages measuring the severity. Comparative findings took a turn to establish that pedestrian severity increased steeply with alcohol impairment and on the contrary, motorcyclist severity elevated steeply in high-speed zones. The drawbacks were poor model specificity implying that deep learning could be used in the future. The implications of the study were practical and pointed out that specific safety plans such as variable speed limit and awareness would be effective. The SVM grid-search tuning of hyperparameters or hyperparameter tuning guaranteed powerful model performance. It was found that the most effective model was RF, and the most vulnerable was the motorcyclists, and the interventions were required in the first place. This is because of its novel linking of feature ranking and partial dependence plots to get subtle but actionable information when it comes to VRU safety policies.

Author[16] To investigate the problem, this paper suggests the ensemble deep learning model that will predict accidents on the road in Bogota, Colombia based on the data about social media

and fall in weather conditions. The model coupled the use of Gated Recurrent Units (GRU) with Convolutional Neural Networks (CNN) in examining spatial-temporal patterns and weather conditions. The study deals with such challenges as data availability, noise in the social media reports, and the non-linear accident type.Information on Waze (May 2018 to June 2019) and weather data were gathered, cleaned of anomalies and converted into a 1 km x 1 km spatial grid. The feature engineering process implied the accident probability calculation and synthetic non-accident reports being added to data so that the dataset could be balanced. The ensemble model takes the time-series data (GRU) and runs it alongside the temporal information (peak hours, cnn) and the effect of rainfall (cnn) and combines the results of these to make the final predictions.The findings demonstrated that the model performed better than baseline algorithms (AdaBoost, Linear Regression) and other deep learning methods as the values of RMSE (0.22) and MSE (0.05) were low. The specificity of the given research was a combination of using social media along with climate data as well as a data augmentation solution, which allowed creating the synthetic non-accident records, enhancing the robustness of the models. The correlations are helpful in preventing accidents by traffic agencies especially in high risk areas such as intersections along the corridors. In future, one can add other variables such as traffic congestion and severity of accidents.

Author[17] The paper investigates the machine learning methods to study the data about traffic accidents in Lipetsk, Russia (20142019) in order to minimize accidents by defining the profiles of drivers and predicting the type of accident. The k-means cluster algorithm is applied to groups of seven homogeneous sets of drivers where the silhouette coefficients and the elbow technique validate the findings such as the most likely occurrence of crashes involving male drivers, age 30-40, having 1 13 years of driving experience. This is the first work that combines the concepts of clustering and predictive modeling in order to target social media campaigns to a specific group of drivers. The potential of machine learning has been explored by testing six different machine learning models, namely, logistic regression, ridge regression, decision trees, random forest, CatBoost, and neural networks, which are needed to determine the type of accidents by using driver features (e.g., age, violations). Logistic and ridge regression obtained the most accuracy (~68%). The research demonstrates an innovative strategy: clustering of the driver portrait alongside the predictive analytics in order to improve road safety campaigns targeting. The limitations are the specificity of regional data, although the method has a way of scaling. The insights are useful to the policymakers in developing quality interventions that are data-informed and will reduce traffic crimes.

Author[18] The objective of the work is to compare the binary logistic regression (BLR) with the hybrid machine learning model support vector machine (SVM) + Whale Optimization Algorithm (WOA), support vector machine (SVM) + Grasshopper Optimization Algorithm (GOA), and support vector machine (SVM) + Multi-Verse Optimizer (MVO) to predict the level of road crash severity (LRCS) in Cosenza, Italy. The study picks 1,600 crash reports (2015-2018) to

measure such factors as the number of vehicles, the type of road, and the speed limit. The hybrid models, specifically SVM-WOA, were able to converge early and were comparable with the accuracy of BLR (65.5% prediction accuracy) but without significant variations in performance. This study is unique in using the societal cost thresholds (42,000 euro) to determine the crash severity, and to combine the economic impact on the crash severity measures in a more common classification. Considerations of sensitivity reflected that vehicles in number were the most sensitive factor and road factors to be least sensitive. The paper sheds light on the prospects of hybrid AI models in road safety planning, recommending future modifications of such models by use of deep learning. The data normalization, k-fold validation, and cosine sensitivity analysis were used as methodologies to verify data. The work plays a role in sustainable mobility through the optimum forecasting of crash severity to make strategic safety interventions.

Author[19] The proposed study aims at predicting incidents of road accidents in Hubballi-Dharwad in India based on deep learning (DL) models into developing better safety regulations regarding roads. They have provided a comparative analysis of 6 DL models namely RNN, LSTM, CNN+LSTM, GRU, Transformer, and MLP based on a 10-years data computed in Smart Road Accident Database. LSTM was the most accurate and recorded the least errors (RMSE: 0.291 yearly, 0.186 monthly; MAPE 6.674 yearly, 5.850 monthly).The methodology included a data preprocessing procedure, train-test split (80:20), and training the model with such validation metrics as MAE, MAPE, and RMSE. This was the only study that has integrated time and climatic variables, which found out that more accidents occur during monsoons because of poor visibility and poor road conditions.Findings indicated that the monthly forecast performed better than the yearly one, which helped policy suggestions both in short term (e.g. speed regulation) and in the long term (e.g. infrastructural upgrades). What makes the study unique is that it has given a detailed comparison of modeling approaches and it also gives special focus to weather-induced distribution of accidents, so that urban planners can readjust to curtail the number of accidents and enhance safety.

Author[20] This article proposes a unique mechanism of predicting the severity of traffic accidents with the Deep Forests algorithm, a machine learning method that is more effective than the existing conventional platforms such as Random Forests or neural networks. The study uses the road safety data provided by the UK that contains 18 variables/features (age of the driver, car type, the weather, and speed limit) in its dataset (2016) Among the crucial procedures, there were data preprocessing (it included cleaning outliers data and normalization, as well as reduction in the imbalanced classes), and feature selection, which was performed based on the Random Forests algorithm to determine the most impactful variable features such as the engine capacity and the hour of the day. Based on the Deep neural networks but with reduced hyperparameters the Deep Forests model is a cascade of ensembles of decision trees. It can adapt automatically to its complexity depending on the level of validation performance which does not require manual adjustment. It has been compared in experiments (with its accuracy compared to those of other

algorithms, DNN, XGBoost, etc.) and has a demonstrated 90.69 percent accuracy and high stability, even with small samples. One thing which was special was k- Cross-validation inside the cascade layers to produce class vectors, and to increase prediction robustness. The results point out the potential of Deep Forests in real-life traffic safety systems, which can be helpful to policymakers in active accident prevention. In the future, it can target the feature engineering and param optimisation to increase the precision.

NOTE: For another 20 research papers the Summary is in the form of a table

S/ No	Name of Paper	Author Name	Year of Publish	Methodology	Limitations	DOI
21	Road Traffic Accident Risk Prediction and Key Factor Identification Framework Based on Explainable Deep Learning	Yulong Pei, Yuhang Wen, Sheng Pan	2024	CNN-BiLSTM-Attention model of the spatiotemporal feature extraction, DeepSHAP of the interpretability and keys factor recognition	Restrictions to US and UK datasets, reliance on data quality, high computational expense	**10.1109/ACCESS.2024.3451522**
22	A Review of Artificial Intelligence and Machine Learning for Incident Detectors in Road Transport Systems	Samuel Olugbade, Stephen Ojo, Agbotiname Lucky Imoize, Joseph Isabona, Mathew O. Alaba	2022	ANN, SVM, IoT, VANET, 5G/6G, and other AI/ML techniques for incident detection are thoroughly reviewed, real-time tracking, predictive maintenance, and traffic management are highlighted.	Theoretical review only, no empirical support, reliance on new technology (5G/6G)	**10.3390/mca27050077**
23	Prediction of roadway accident frequencies: Count regressions versus machine learning models	H. Nassiri, P. Najaf, A. Mohamadian Amiri	2014	Support Vector Machine (SVM), Back-Propagation Neural Network (BPNN), Negative Binomial Regression (NBR), and Zero Inflated Negative Binomial Regression (ZINBR)	Dataset restricted to a single city (Mashhad, Iran), brief time frame (2007), and challenges interpreting machine learning models (black-box	**10.24200/sci.2014.1630**

					nature)	
24	Predicting Car Accident Severity in Northwest Ethiopia: A Machine Learning Approach Leveraging Driver, Environmental, and Road Conditions	Abraham Keffale Mengistu, Andualem Enyew Gedefaw, Nebebe Demis Baykemagn, Agmasie Damtew Walle, Tirualem Zeleke Yehuala, Meron Asmamaw Alemayehu, Mengistu Abebe Messelu, Bayou Tilahun Assaye	2025	Logistic Regression, Decision Trees, Random Forests, Gradient Boosting, XGBoost, LightGBM, SVM, KNN, Neural Networks, Naive Bayes	Dependency on historical data from a single region (Northwest Ethiopia) from 2018–2023, The sample size is small (2000 records), Interpretability is diminished by ML models' black-box nature, Class imbalance-related bias (SMOTE applied).	**10.1101/2025.01.12.25320441**
25	Dynamic prediction of traffic incident duration on urban expressways: A deep learning approach based on LSTM and MLP	Weiwei Zhu, Jinglin Wu, Ting Fu, Junhua Wang, Jie Zhang, Qiangqiang Shangguan	2021	hybrid deep learning framework that combines MLP (for incident-related factors) and LSTM (for dynamic traffic flow data). The 30-minute time window, traffic volume, and speed inputs are important parameters. Metrics: accuracy (80.7–85.7%), AUC (0.86–0.94).	Deep learning models have limited interpretability, A small dataset consisting of 391 incidents from the Zhonghuan Expressway in Shanghai, Does not include variables such as injuries/casualties or precise location information, Disregards correlations	**10.1108/JICV-03-2021-0004**

					between variables.	
26	The Prediction of Road-Accident Risk through Data Mining: A Case Study from Setubal, Portugal	David Dias, José Silvestre Silva, Alexandre Bernardino	2023	Risk classification (low, medium, and high), feature selection (Cramer's V, Kruskal-Wallis), and data mining (kNN, regression, decision trees, and neural networks)	No real-time mobility data, reliance on weather forecasts, and limited to Setubal district data (2019–2021)	**10.3390/informatics10010017**
27	Enhancing prediction and analysis of UK road traffic accident severity using AI: Integration of machine learning, econometric techniques, and time series forecasting in public health research	Md Abu Sufian, Jayasree Varadarajan, Mingbo Niu	2024	Machine learning (Random Forest, XGBoost), Econometric methods (VAR, GMM), Forecasting time series (ARIMA), Explainable AI (SHAP), Multivariate statistical analysis.	Reliance on government-reported data; exclusion of unreported or minor accidents; restricted to UK data (2019) High H2O AutoML computation time	**10.1016/j.heliyon.2024.e28547**
28	Towards Benchmark Datasets for Machine Learning Based Website Phishing Detection: An Experimental Study	Abdelhakim Hannousse, Salima Yainouche	2020	Dataset Construction, Feature Selection, Experiments, Evaluation	Dataset Dependency, Feature Effectiveness, External Service Delays, Limited Deep Learning Exploration	**10.17632/c2gw7ly2t4-2**
29	Prediction of Accident Risk Levels in Traffic Accidents Using Deep Learning	Cristian Arciniegas-Ayala, Pablo Marcillo, Ángel Leonardo Valdivieso	2024	Developed the PoliDriving dataset (2634 samples, 23 features).Used Pearson correlation for	Dataset-dependent results. High computational time for deep	**10.3390/app14146248**

	and Radial Basis Function Neural Networks Applied to a Dataset with Information on Driving Events	Caraguay, Myriam Hernández-Alvarez		feature selection.Compared CNN, CNN-RF, RBF (GPC/SVC), and MLP models.Evaluated using accuracy, specificity, sensitivity, and processing time.	learning models.Required undersampling due to imbalanced data. Limited exploration of dynamic dataset integration.	
30	Traffic Accident Severity Prediction Based on Random Forest	Miaomiao Yan, Yindong Shen	2022	Hybrid model integrating Random Forest (RF) and Bayesian Optimization (BO) for hyperparameter tuning. Uses relative importance and partial dependence plots for interpretability.	Dataset limited to Montgomery County, Pennsylvania (may lack generalizability), Focus on urban roads; results may not apply to highways.Computational constraints reduced dataset size.	**10.3390/su14031729**
31	Predicting Traffic Accidents and Their Injury Severities Using Machine Learning Techniques	H. L. Gururaj, V. Janhavi, U. Tanuja, Francesco Flamini, B. C. Soundarya, Vinayakumar Ravi	2022	Unsupervised Learning using Apriori, Apriori TID, SFIT, and ECLAT algorithms for association rule mining to predict injury severity patterns	Requires historical unlabeled datasets; performance depends on dataset quality and diversity	**10.2495/TDI-V6-N4-363-377**
32	Advanced accident prediction models and impacts assessment	Fabio Galatioto et al.	2018	Developed multiple accident prediction models integrated within a web-based road-safety simulation platform	May rely on simulated inputs, limited validation on real-world accident data	**10.1049/iet-its.2018.5218**

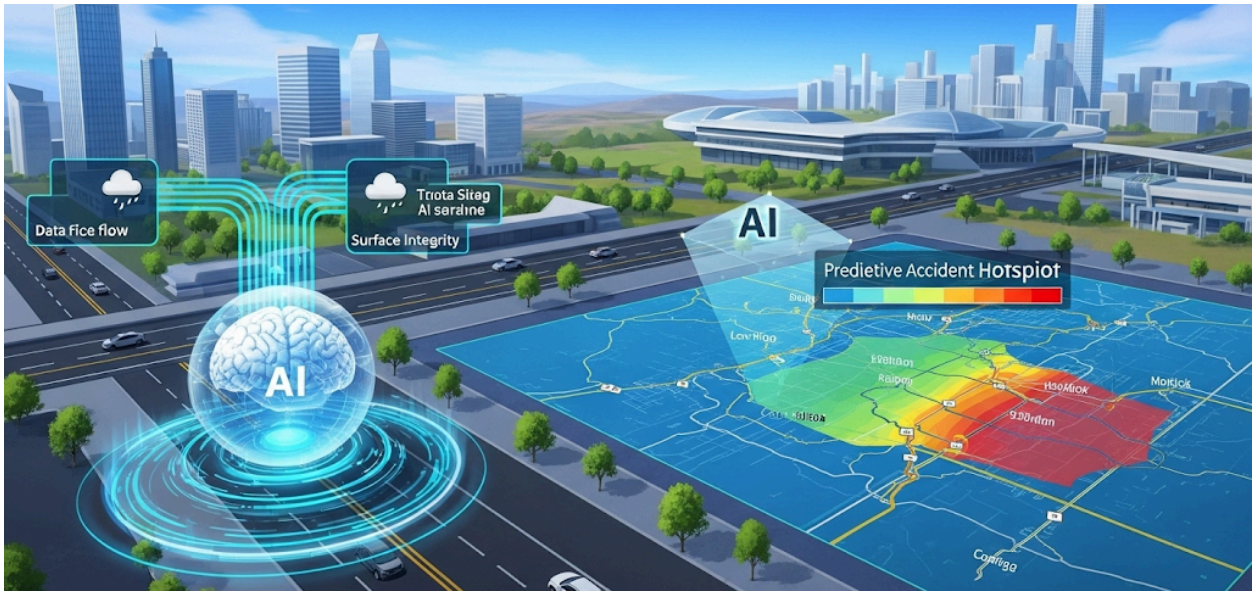
33	Using Machine Learning in Predicting the Impact of Meteorological Parameters on Traffic Incidents	Aleksandar Aleksić, Milan Ranđelović, Dragan Ranđelović	2023	proposed a novel ensemble machine learning model that combines logistic regression, classification (e.g., Naive Bayes, J48), and feature selection (filter + wrapper methods); it is then implemented in a two-agent multi-agent system for real-time alerts and prediction..	results based on a case study from Niš, Serbia (1992–2009); restricted to meteorological data; lacks wider external validation	**10.3390/math11020479**
34	Predicting Maritime Accident Risk Using Automated Machine Learning	Ziaul Haque Munim, Michael André Sørli, Hyungju Kim, Ilan Alon	2024	Trained and tested 29 ML models by applying AutoML on the platform of DataRobot. The best-performing one was the Light Gradient Boosted Trees Classifier. Information on 9,025 ship accidents following 40 years (Norwegian waters).	Only Norwegian data can be used; not accurate position of vessels; no weather or sensor information; and not tested in other than Norwegian waters.	**10.1016/j.ress.2024.110148**
35	A novel approach to predict the traffic accident assistance based on deep learning	José F. Vicent, Manuel Curado, José L. Oliver, Luis Pérez-Sala	2025	The three-phase DL framework includes the following: (1) Data pre-processing (cleaning, transformation, area filtering, resampling); (2) Post-processing with boosting and genetic algorithm, calculating feature importance; (3) CNN-2D model which predicts the requirements of medical help by using	The extent of generalizability can change city to city; is dependent on how well resampling and input transformation is carried out; no external country datasets	**10.1007/s00521-024-10939-z**

				a simplified restructured matrix-based datasets (with 6 UK cities tested).		
36	Traffic Accident Prediction Based on Multivariable Grey Model	Wei Li, Xujian Zhao, Shiyu Liu	2020	Used grey correlation analysis to identify key influencing factors, then applied Multivariable Grey Model (MGM(1,N)) for prediction. Model optimized via variable selection and validated against real accident data from China.	Chinese data only, few variables, the results could not be applicable to other regions or complicated traffic conditions.	**10.3390/info11040184**
37	Prediction in Traffic Accident Duration Based on Heterogeneous Ensemble Learning	Yuexu Zhao, Wei Deng	2022	Constructed a heterogeneous ensemble of XGBoost, LightGBM, CatBoost and ElasticNet models on Kaggle data provided on 2016-2020 real data of incidents across the U.S. Meta learning: SHAP analysis, Applied feature engineering, imputation (K-Nearest neighbor, Random Forest) stacking in predicting the duration of accidents at the initial phases.	Is sensitive to only early incident data (limited real time data); the sophistication of the model; it might fail to generalize to other world-wide data sets or conditions as live stream.	**10.1080/08839514.2021.2018643**
38	Prediction of Traffic Accident Casualties Using Machine Learning: For Seoul Public Data	Myung-woo Nam, Doo-Seo Park, Young-Jun Jang, Hong-Chul Lee	2021	With the data of the traffic accidents in Seoul and weather data of 2015-2019, tree-based ML models (DecisionTree, RandomForest, XGBoost, LightGBM)	Datasets consisted solely of fatal accidents; imbalanced casualty values; the only ML used	Not available (Korean Computer Information Society, 2021)

				were used. The tuning of the models was done through the GridSearch + 5-fold CV. LightGBM provided the most accuracy (the lowest MAPE).	is tree-based; future research implies an extended number of different algorithms and datasets.	
39	Detection of Accident Situation by Machine Learning Methods Using Traffic Announcements: The Case of Metropol Istanbul	Eren Dağlı, Mustafa Büber, Yavuz Selim Taşpınar	2022	Applied a collection of 136,964 traffic announcements in Istanbul to determine the categorization of accident scenarios over Artificial Neural Networks (ANN), k-Nearest Neighbors (kNN), and Decision Trees (DT). AUC, ROC analyzed, F1-score with precision.	ANN had the best performance in accuracy (92.1) and was the slowest in training; DT the fastest in testing and the worst accuracy (89.8); the model low-data classes performed poorly because it was skewed.	**10.18100/ijamec.1145293**
40	Collision Cross Section Prediction with Molecular Fingerprint Using Machine Learning	Fan Yang, Denice van Herwerden, Hugues Preud'homme, Saer Samanipour	2022	Trained two random forest ML models of predicting collision cross section (CCS) using topological molecular fingerprints. One of them based the prediction on the classes; the other one predicted directly. A sample of 13, 324 CCS values was utilised.	Accuracy of non-routine/complex molecules; bias in the dataset; a few classes of structural type had a small number of examples; omitted adduct ions.	**10.3390/molecules27196424**

METHODOLOGY

Machine learning (ML) has become an important instrument that helps to predict road accidents with the help of large volumes of historical and timely data. Unlike classical statistical models, ML algorithms have the capability of automatically identifying complex patterns and relations present in the data hence they would suit such dynamic systems like traffic systems. The procedure starts by the gathering of a variety of data such as previous accidents, weather, traffic volume, and driver behaviors. This information is then filtered, and fed into predictive units which rate the degree of risk or a prediction of the occurrence of an accident.



Adaptability is one of the primary resources of machine learning in this sphere. As additional data is made available, it is possible to retrain models to obtain better accuracy. As a case in point, a system consisting of five years of accident data can further improve its forecast once it is updated with fresh data, say modifications in the road infrastructure or traffic rules. Also, ML models can manage real-time input such as real-time weather conditions or traffic congestion reports and repeat on the spot risk assessment. It is important in applications such as smart traffic management systems which could give warnings to drivers or change signal lights to avoid accidents.

The second value is the possibility to discover previously unknown correlations which is not possible to reveal by using manual direction. In another example, a model may find that fog-prone weather conditions, rush-hour traffic jams and twisters on the roads result in a high rate of accidents. This kind of insight can allow the authorities to develop specific safety interventions, such as placing warning signs or putting up more patrols in danger areas.

Data Collection and Preprocessing :

1. **Data Sourcing**:
   Effective prediction of accidents is dependent on high quality multi-source data.

   i) **Historical accident records**: Accident history records contain basic information such as time, place, severity and causes of past accidents.

   ii) **Environmental factors**: Other environmental data like weather conditions (rain, snow, fog) or quality of the roads (wet, icy conditions, potholes) play a decisive role as they affect the level of driving safety directly.

   iii) **Traffic data**: Traffic-related data, i.e.: the number of vehicles per type, average speed, and peak hourly characteristics are used to recognize the risks linked to congestion.

   iv) **Driver-specific factor**s: Additional refinements in the predictions are through driver-specific factors such as age, experience and history of violations that factor human error.

2. **Data Cleaning**

   Raw data are usually in inconsistent or missing value or outlier forms that can distort predictions. There are methods to clean up data, such as ignoring missing pieces of information through mean imputation (replacing the missing data with the average), or more sophisticated such as Multiple Imputation by Chained Equations (MICE) which takes into account the relationship between the variables, which is used to smooth the data as much as possible. Outliers, like unrealistic speed measurements are eliminated so as not to distort the training of the models. Another important step will be normalization where all variables (e.g., temperature, speed limits) are put on a similar scale. As an example, when speed (0 average up to 100 mph) and temperature (average to 40C down to -10C) values are converted to be on a scale of 0 to 1, the larger range of the variable does not overpower the model above the other variable with a smaller range.

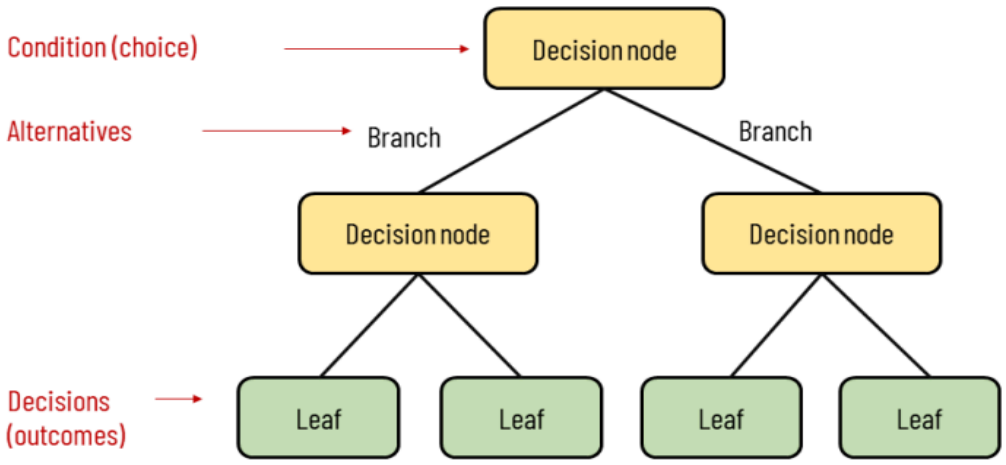
3. **Feature Selection**

   Every data feature does not have the same exact role in predicting. SHAP (Shapley Additive Explanations) and other feature selection methods put variables in order according to their influence. As an illustration, one of the Research Papers [7] listed speed limit and time of crash as the strongest predictors, whereas another [15] found out that alcohol impairment raised the proportion of pedestrian fatalities more substantially than would be the case. With the attention paid to the most influential features, models are more efficient and interpretable.

Machine Learning Models for Accident Prediction

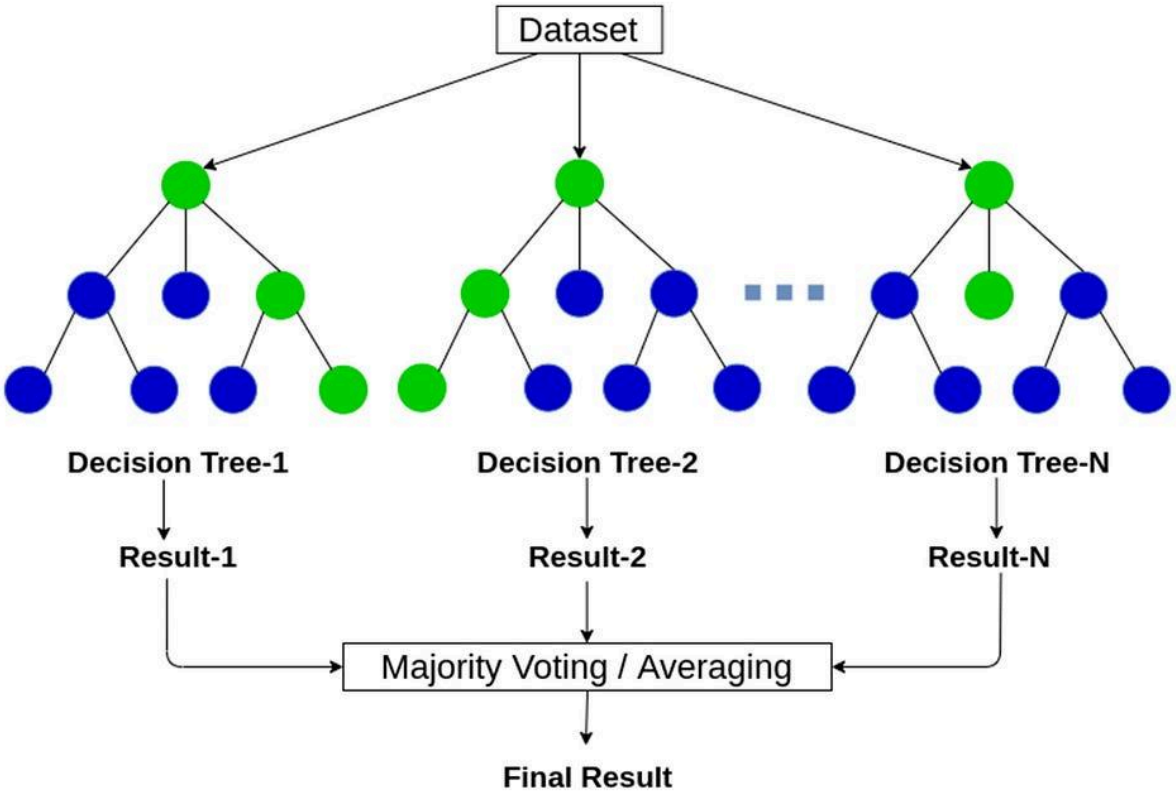
1. **Decision Trees**

   Decision trees resemble flow charts which are used to give predictions using a series of yes or no questions on the basis of the data fed into it. Think of attempting to make a guess as to whether a road accident could occur or not--the tree begins with a large question (e.g, "Was it raining?") and fans out according to its answer (e.g, "Yes?" Check speed Limit." / "No? Check the traffic density("Look at traffic density."). One decision eliminates one possible outcome after another until the last prediction is concluded (e.g. High risk or Low risk). They can be easily comprehended since they resemble the manner in which a human makes decisions by simplifying their complex problems into simple processes. When it comes to road accident prediction, they are these decision points, as, based on the weather, class of roads, or behavior of the drivers, the model can make an approximation of the likelihood of an accident.



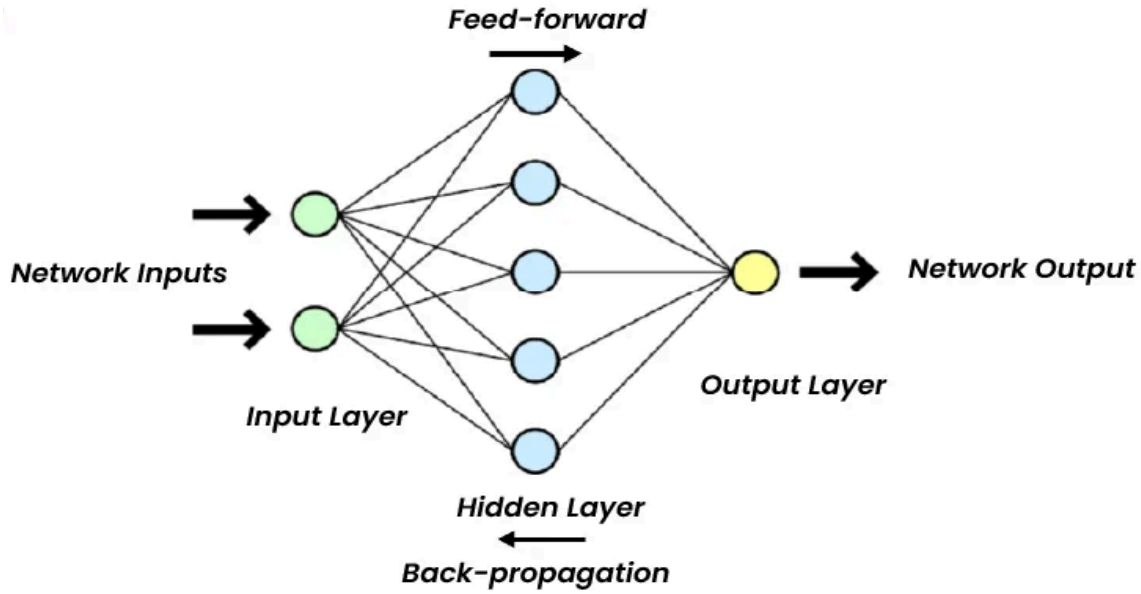
2. **Random Forest**

Random Forest can be described as a joint effort of a group of decision trees to make more intelligent forecasts. Rather than having only the one flowchart (a single decision tree) it generates numerous trees, each exploiting the various parts of the data or concentrating on different variables (e.g. weather, speed, type of road). Once all trees have given their vote on whether there will be an accident or not, the last prediction is given by the majority vote-this is similar to obtaining the average of the views of many specialists. This increases the accuracy of random forests and reduces the possibilities of errors since they compensate each other. This is useful particularly in prediction of road accidents because it incorporates several viewpoints and no individual aspect can influence the outcome to a great extent.



3. **Neural Networks**

Neural Networks are an intelligent digital brain that works like the brain of a person. Think of a network of minute decision-makers (referred to as neurons) which operate in a stratum, where each stratum sees to it the different hints of data, such as pushing models in weather, traffic or road condition. Simple clues that may be picked up by the first level may be put together by deeper levels to interpret higher levels of relations (e.g. Rain + night + sharp turn = high risk of accidents). The more it learns, the more accurately it can predict accidents. Unlike (an) decision tree or random forests, neural networks tolerate messy real-world data very effectively so in cases where we want to make a tricky prediction they are extremely powerful (at least for some specific case). They work like a black box though, which in other cases makes them less useful (from an explainability perspective).



Hybrid Models

Hybrid Models are a kind of dream team of machine learning, as diverse algorithms (such as decision trees, random forests and neural networks) combine their forces to make predictions more intelligent and precise. Rather than relying on a single approach, a hybrid model is able to adopt the positive aspects of each by, in the case of accident data, integrating neural network capabilities on identifying complex patterns with another technique, a random forest, performing a second check on the data reliability. It is a case of two different heads with one being quick and instinctive and the other slow and

methodical. In prediction of road accidents, this implies the start of more accurate predictions since the model can deal equally well with both simple factors and more complicated hidden trends, such as the sudden weather shift. They are more difficult to construct, although the reward is usually more precise and reliable forecasts.

Model Validation and Deployment

Ensuring Real-World Reliability

Before deployment, models can undergo rigorous validation to ensure they perform accurately in real traffic conditions. Cross-validation techniques can be employed that repeatedly test the models on different subsets of data, simulating various scenarios they might encounter. This helps identify and address any weaknesses while maintaining consistent performance across diverse situations.

Practical Implementation Strategies

Successful models can be transitioned from testing to real-world application through phased deployment. Initial pilot programs in select urban areas can allow monitoring performance while minimizing risk. These controlled implementations help refine the system's responsiveness to actual traffic patterns, weather conditions, and driver behaviors that lab testing can't fully replicate.

Continuous Improvement Cycle

Deployed models can enter an ongoing optimization process where they continuously learn from new accident data and near-miss incidents. This living system will adapt to changing road conditions, vehicle technologies, and traffic regulations, ensuring predictions remain relevant over time. Regular performance audits and driver feedback mechanisms can create a responsive safety ecosystem that evolves with real-world needs.

ANALYSIS

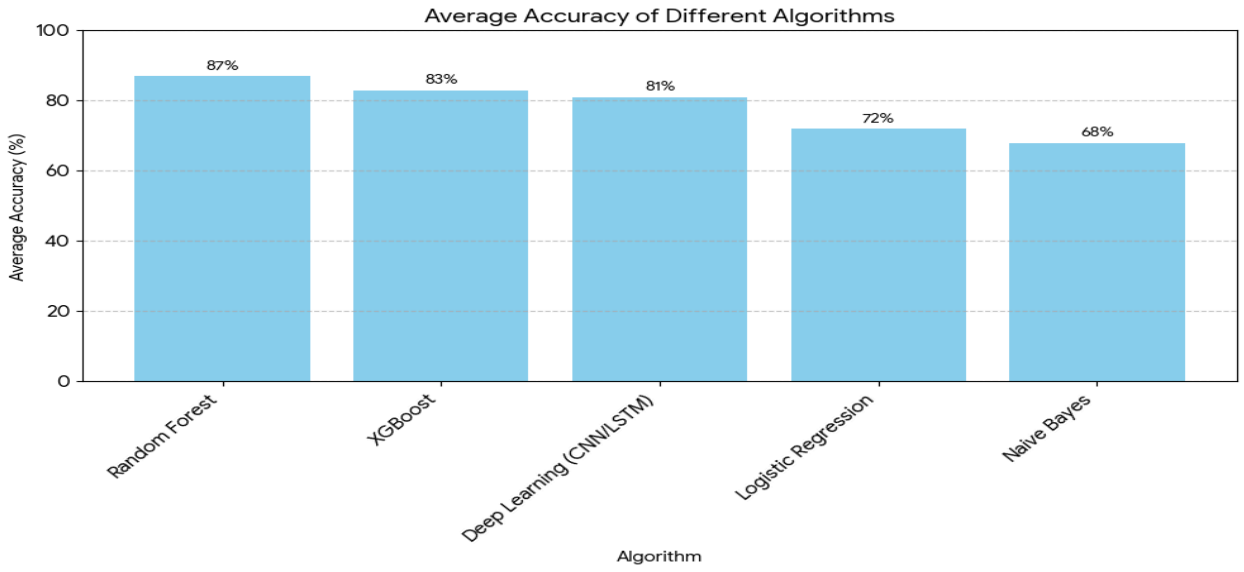
These research papers all point to the increased use of machine learning (ML) and deep learning (DL) models to forecast the severity of road accidents and the factors that contribute to the same. Random forest (RF) will be found as the most common and successful algorithm, and will yield high accuracy (e.g., 85-97% in Papers [1], [5], [6], and [10]) because it enables it to deal with an unbalanced dataset and analyze the importance of features. It is, of course, better with hybrid models like RFCNN (Paper [7]) or ensemble methods (Paper [22], Paper [37]) which combine the power of several algorithms, e.g. RF with convolutional neural networks (CNNs) or XGBoost. Such developments deal with shortcomings such as imbalanced representation of classes (e.g., with SMOTE in Paper [2]) and even the noise in the data, but they continue facing issues when it comes to forecasting rare-but-deadly accidents (Paper [9]).

Changes in geography and the situation strongly affect the behavior of models. To mention only some cases, research based on the UK (Papers [1], [20]) inclines to weather and road situations, whereas research in India (Papers [6], [19]) is on driver behavior and infrastructure. Integration of spatial-temporal data observed in both Papers [16] (Colombia) and [19] (India) increases the accuracy of prediction by including real time conditions such as social media signals or monsoon effects. Nevertheless, the scope of the information collected in the regions is limited (e.g., Paper [17] focuses on Lipetsk, Russia) to prevent generalization, which is another reason to tune the model on a local basis.

The question of interpretability is attached with utmost importance. There is better transparency with SHAP (Paper [2]) and partial dependence plots (Paper [15]), but simpler models such as logistic regression (Paper [13]) perform better when it comes to interpretability under policymaking than ML. Deployment is complicated further by such ethical issues as data bias (Paper [5]) and privacy (Paper [28]). Applications under the themes of web-based application (Papers [5], [26]) and a real-time alert (Paper [33]) essentially fill in the gap between a research and an implementable solution, but may not work well in non-static conditions due to use of historical data (Paper [25]).

Future trends consist of utilizing multi-domain data (e.g., traffic data, weather, IoT, etc.) as well as exotic methods such as Deep Forests (Paper [20]) or transformers (Paper [19]). Paper [7] and Paper [16] promote hybrid AI models to find a compromise between the accuracy and quasi-computational performance, whereas Papers [12] and [14] focus on clustering and association rule mining as an objective of pattern detection. Policymakers are called to consider implementing implementable knowledge, like speed limits (Paper [15]) or high-risk areas (Paper [14]) in order to pursue international safety strategies such as Road to Zero (Paper [2]).

Aspect	Insights	Example Papers
Top Algorithms	Random Forest dominates; hybrid models (RFCNN, SVM-WOA) show higher accuracy.	1, 5, 7, 18
Critical Features	Weather, road type, speed limits, driver age, and multi-vehicle collisions are recurrent predictors.	2, 9, 15, 20
Challenges	Class imbalance, regional bias, interpretability trade-offs, and real-time data integration.	6, 13, 25, 33
Innovations	SHAP analysis, Deep Forests, social media data fusion, and ensemble learning.	2, 16, 20, 37
Policy Applications	Hotspot identification, dynamic speed limits, and targeted safety campaigns for high-risk demographics.	7, 14, 17, 19



CHALLENGES AND LIMITATIONS

Development of machine learning in the field of road accident prediction has been a revolution, however major obstacles exist, which affect its practical performance in the real world. These complexities include the challenges of data quality, the performance of the models used, implementation and ethical issues, which are complex and challenge researchers and practitioners. Such issues have to be solved to get the maximum out of AI in the context of road safety.

1. **Data Quality and Availability Issues**

   Among the most basic problems is the quality of accident data and its availability. Numerous papers depend on government-reported data that usually have incomplete reports or bias in reporting data. An example of this is that small scale accidents will not be accurately created and large scale crashes will be overrepresented, forming biased training sets. Even the regional variations make the situation more complicated - the model that was developed on the UK network of roads might fail when exposed to the Indian traffic conditions because of different infrastructure, traffic rules and driver application enforcement. The lack of information e.g. the weather conditions are not reported, or partial information of drivers needs to be included, necessitates excessive pre-processing that may end up having its own biases.

2. **Class Imbalance and Rare Event Prediction**

   The nature of the severity of accidents itself poses great difficulties to the class imbalances which are a problem to machine learning models. Although the minor accidents are rather frequent, the most serious crashes that result in deaths or bad injuries are rather infrequent. This disparity causes the models to have high accuracies by only predicting the most of the majority classes and will have low accuracies in the important cases of the minority. Various studies identified this problem, and this paper did not record above 19.8 percent recall on the fatal accidents, although overall accuracy was good. This is an issue that is poorly handled by traditional sampling methodology, and needs more advanced techniques such as synthetic minority oversampling or loss functions which are specialized.

3. **Interpretability Versus Accuracy Trade-off**

   Models which are increasingly complex in the effort to produce improved predictive performance tend to perish at the altar of interpretability, producing a big setback to the application on the field of relevance. As much as the deep learning ensemble can reach as high as 90 percent accuracy, its opacity offers little insight available to the traffic authorities as to the predictions made by the model and, therefore, they do not trust it. Conversely, more straightforward models such as logistic regression do not give many practical insights regarding risk factors, although they usually are less predictive. It is the conflict between accuracy and explainability that is a principal issue in actual usage of these systems, especially when human lives and policymaking are involved.

4. **Real-Time Implementation Challenges**

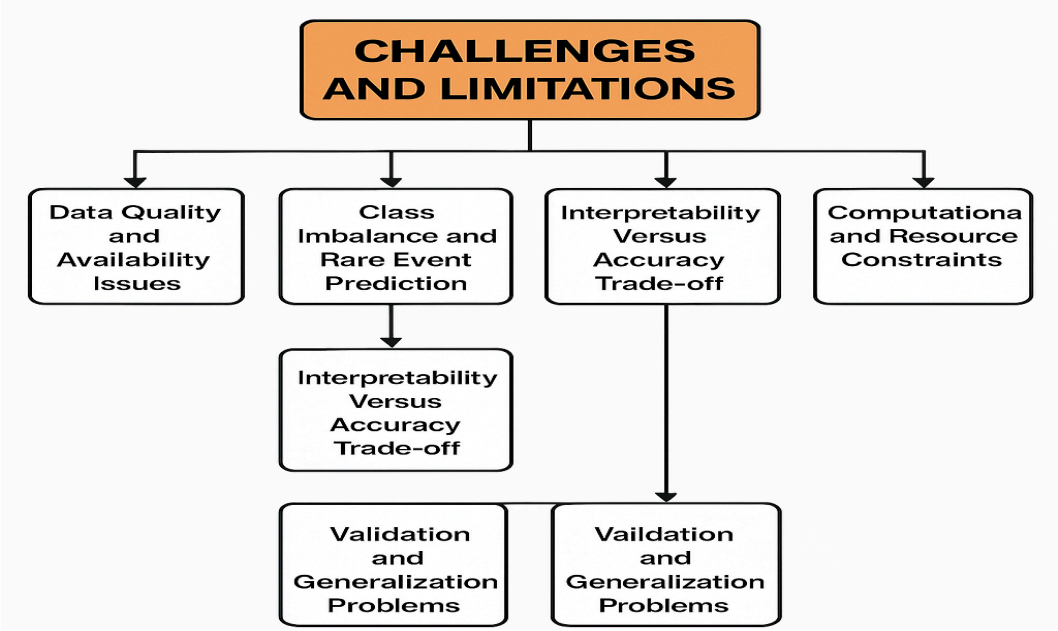
   Some of these problems include the difficulty in maintaining the changing road conditions in a major challenge related to the accident prediction systems. Traffic in the real world cannot be controlled, like it is in a lab - any surprise shower, unexpected work zones, or concert venue can turn driving behavior completely in seconds. Although using advanced sensors and active data feeds would be helpful, most of the cities are unable to provide a sufficient technology base and these systems are forced to rely on old data which cannot be able to reflect the constantly changing realities on the roads nowadays.

5. **Computational and Resource Constraints**

   The sophisticated machine learning algorithms require high amounts of computational resources thereby limiting their feasibility. Deep learning models based on accidents data involve strenuous processing power and energy requirements. These resource necessities make many of the promising strategies inapplicable to any widespread use, especially in developing areas when the need for road safety could be highest and resources for computation are minimal. Also, models trained to work under conditions in a particular domain (such as city intersections) often do not apply well to other contexts (such as countryside highways), and need costly retraining, or adaptation.

6. Ethical and Privacy Considerations

Ethical issue of using sensitive personal information in the prediction of accidents is also a concerning topic. The data usually contains personal information about drivers (including demographic data and medical records), which needs to be properly secured. The issue of algorithmic bias is another one of its critical concerns, where it is possible that models can incidentally discriminate against one of the demographic groups when training data reifies the biases that exist in the society itself. The problems are made especially critical when the effects of prediction systems affect insurance rates, traffic enforcement or funding of infrastructures. It is a constant challenge to establish ethical frameworks of AI applications in transportation safety.



7. Validation and Generalization Problems

A large number of research studies struggle to legitimize their models in the various geographies and horizons. An effective model in a specific region can fail in other regions because of the variation in the road systems, patterns of driving people and reporting of data. A big problem is that there is no coherence in the way performance is measured, accuracy is used in some papers and precision/recall in others. This discrepancy provokes the problem of comparing results and realizing which models are really more effective in practical conditions. This in effect makes it hard to design general solutions

CONCLUSION

Machine learning has clearly revolutionized road safety research and has introduced very strong tools to anticipate or prevent road accidents. The way through these studies shows a tremendous improvement on how the models have reached a level of impressive accuracy in understanding the complex pattern of the historical data. But behind these feats of technology are very human obstacles that serve to warn us that technology in and of itself is not a miracle worker. The actual roads do not play nice with the laboratory data - they are tweaked areas with some randomness where sudden storms, lost concentration of drivers, and old infrastructure plot against ideal forecasting.

These thresholds that we have revealed are testimonies to other problems to our implementation of these systems. Pristine algorithms are doomed to failure when fed with incomplete or biased data, missing exactly those accidents that we would most like to avert. Even very accurate models have problems with real-time adaptation because of how to explain the moment when a road slicks with rain or the movement of traffic flows is altered by a festival. These are no mere technical glitches - they indicate the difference between experimental simulations and the messy reality of transportation systems.

Moral issues follow all the innovations. That is, when risk is determined by an algorithm, who monitors that it will not disproportionately focus on particular neighborhoods or groups of drivers? What is the trade off between privacy and the necessity of extensive data? Such questions do not have simple answers and shunning them can result in creating systems that contribute to solving mathematical problems but pose problems for society. The most advanced prediction tool will never be of much use, unless it is reliable and comprehensible by the communities in question.

The situation is complicated additionally by implementation issues. Government agencies stretched to their limits have no money to buy sensors and servers to power real-time systems, whereas cities with cash flow constraints have little or no money to buy new sensors and servers. The long-term disadvantages of the new technologies are that these technologies may often meet

the most serious resistance in an environment where they could be most useful. It is not the matter of processing power, it is the matter of political will, budgets and institutional inertia.

Still the hope is too great to lose. One point of better accuracy is equal to a number of lives which could have been saved and injuries averted. The point is to cool down our expectations - we need to think of these tools as supplementary to human judgment, not a substitute for it. They are most effective in augmenting the knowledge of traffic designers, oversight of first responders, and ordinary judgment by drivers.

Future development will be based on inter-disciplinary cooperation. Transportation experts are required to make computer scientists base their models on reality. Policies should be drawn with clear explanations of how these systems operate. Societies should be transparent about the nature of prediction making and utility. It is by this united working forward that only then we can construct systems that will be of the most technical sophistication added to the social responsibility as well.

As the studies progress, one should focus on enhancing real-time flexibility, explainable AI, and the gap in untapped areas. The aim should not be perfect prediction, which can never be achieved, but instead we should develop a system that learns day by day and adapts over time with the roads that it is being tracked. This implies accepting that there exists doubt, but lowering it gradually.

Finally, this work does matter since streets are not the flows of data, but the arteries of our communities. Each accident averted is a family that stays intact, a commercial aircraft worker that makes his or her way home. These technologies may get measured by their score of accuracy but the best measure of them will be the moments of silence, the almost-collisions that never occur, the threats that are caught in time. That is the hope and the potential it has to transform the world on a human level behind all these algorithms and these datasets, and why in spite of the problems, this work is so important.